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Artificial Intelligent

1. A smart house automation system is being designed to temperature and security based on user behavior and Environmental Condition. The system must respond to realtime inputs such as Motion detection, tempurature changes and time of day while also learning user performance over time. In this Context, explain the different type of Intelligent agent (simple reflex, model based, goal based and utilize based agent) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for Ensuring Comfort Energy deficieney and Security.

Ans:-

Introduction

An Intelligent agent is a system that percieves its Environment through sensor and take action through actuators to achieve specific objectives

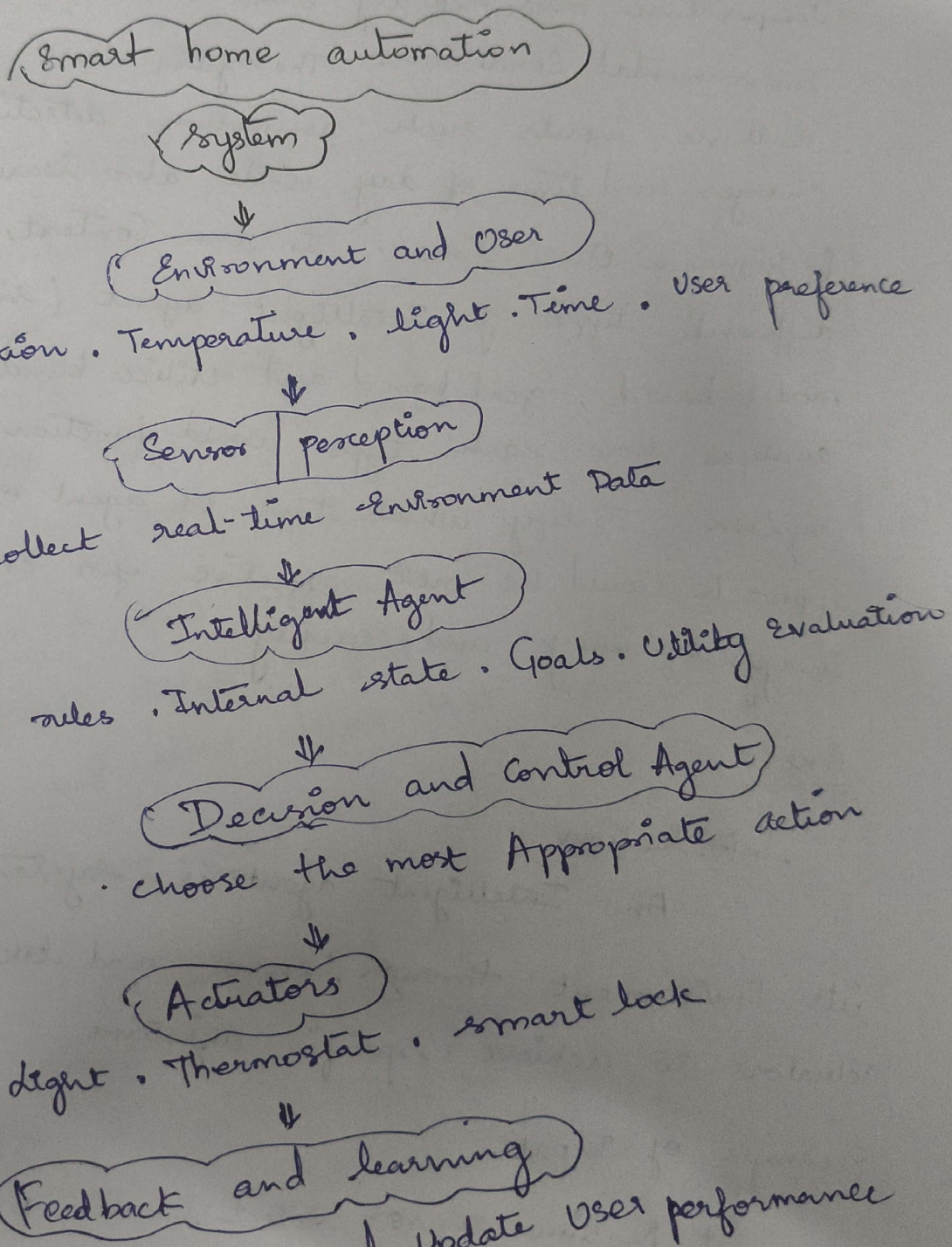
Example of Input

* Motion sensor detect human presence
* Temperature sensor Measure room temperature
* Light sensor Measure Environmental Brightness

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Example of action

* switch digits ON/OFF
* Adjust the air Conditioner or Thermostat
* lock doors and activate alarms.



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* Type Of Intelligent agent

i) simple reflex agent :-

It select an action based only on the current percept, using predefined condition-action rules. It does not remember previous situation

B. Model - Based Reflex agent :-

A model-based agent maintains an Internal state represents relevent information about the Environment. It updates this state using current sensor input and knowledge of how the Environment changes.

C) Goal- based Agent :-

A Goal Based agent chooses actions according to a desired target or goal. It Consider possible action and select those that help achieve the goal.

D) Utility Based Agent :-

A utility based agent Evaluates possible actions using a utility function, which assigns a value to outcome based on how desirable they are. It selects an action that Maximum Expected utility while Considering Uncertainity.

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2. A smart robot is placed in an Unknown maze and must find a path to the Exist without prior knowledge of the layout. Explain how uninformed search strategies like Uniform cost search and Interative deeping search (IDS) can help solve this problem. Discuss the Advantages and disadvantage of these algorithm in Terms of time and space Complexity. Relate the problem to different type of Intelligent agent and Explain how agent design affects decision-Making. suggest which Combination of agent type and search strategy would be most effective and Justify your choice.

Introduction :-

A robot placed in an Unknown maze must find a path from its starting position to the Exit without prior knowledge of the Maze layout.

The two uninformed search strategies Considered are

1. Uniform Cost search (UCS) ⇒ Expands the path with the lowest accumulated Cost.
2. Iterative deepening search (IDS) – Reapeatedly Performs depth limited search with Increasing Depth limit.

Uniformed search algorithm do not use heuristic Estimate of the distance to the goal.

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Maze solving system

Robot Enters Unknown Maze

↓

Sense surrounding

detects walls, open paths and possible moves

↓

Update Internal Maze Map

↓

Apply a search Algorithm

UCS (or) IDS

↓

Is the Exit found?

↓

No

Explore another path, Update the map and search again

Yes

Reconstruct and follow the path to the Exit.

Conclusion:-

UCS and IDS are both useful Uninformed search algorithm, but they serve different needs. UCS is perfered when finding the lowest cost path is important whereas IDS is perferred when Memory is Scarce and all Movement have Equal Cost.

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